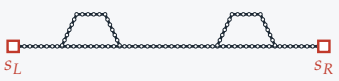
Cross-section analysis: once per cross-section (Sections 2.1, 2.2)

1 Cell



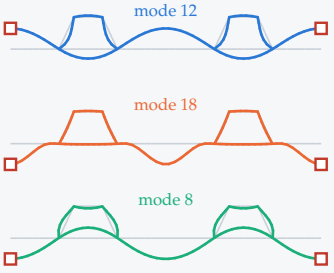
92 nodes; displacements, rotations and warping at the nodes (Eq. (8))

2 Modes, side edges free



families G, D, L, T, S, M (Section 2.1.3)
e.g. rigid rotation: $s_L \neq s_R$, **not periodic**

3 Periodic modes (new)

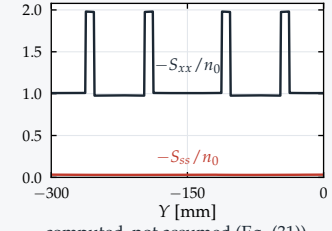


every mode: $s_L = s_R$ (Eqs. (12)–(14))

displacement field $= \sum_k$ *periodic mode* $k\,(s) \times$ *amplitude* $\phi_k(x)$ (Eq. (1)) $\Rightarrow$ periodic for any amplitudes (Proposition 1)

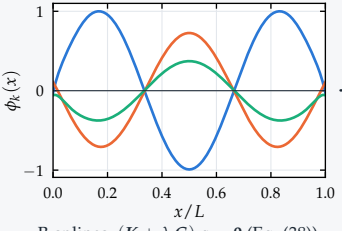
Member analysis: for every length and end condition (Section 2.3)

4 Pre-buckling state



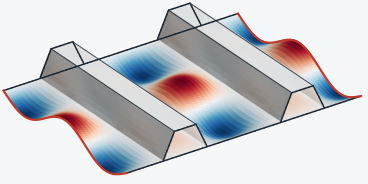
computed, not assumed (Eq. (31))

5 Ritz solution



B-splines; $(\boldsymbol{K} + \lambda\, \boldsymbol{G})\, \boldsymbol{a} = \boldsymbol{0}$ (Eq. (38))

6 Critical load and mode



$N_{cr} = 32.26$ N/mm; mode families (Eq. (41))

Example: Unsym cell, $b_s = 150$ mm, $\beta = 1.5$, SS (Figures 6 and 7). Modes 12, 18, 8 are its dominant hierarchical modes (91.5 % of the strain energy).